

Calculate exact formula for $\int_{\Omega} J_3 v_{\text{CR}}$

Definitions:

General definition from dissertation

$(J_1 v_{\text{CR}})(z) := |\mathcal{T}(z)|^{-1} \sum_{T \in \mathcal{T}(z)} v_{\text{CR}}|_T(z)$ and linear interpolation

$$b_F|_T := \frac{(2n-1)!}{(n-1)!} \prod_{P \in \mathcal{V}(F)} \lambda_P$$

$$J_2 v_{\text{CR}} := J_1 v_{\text{CR}} + \sum_{F \in \mathcal{F}(\Omega)} \left(\int_F (v_{\text{CR}} - J_1 v_{\text{CR}}) \, ds \right) b_F$$

$$\Phi_{T,P} := \frac{1}{\sqrt{|T|}} (\alpha - \beta \lambda_P)$$

$$b_T := (n+1)^{n+1} \prod_{P \in \mathcal{V}(T)} \lambda_P$$

$$J_3 v_{\text{CR}} := J_2 v_{\text{CR}} + \sum_{T \in \mathcal{T}} \sum_{P \in \mathcal{V}(T)} \left(\int_T (v_{\text{CR}} - J_2 v_{\text{CR}}) \Phi_{T,P} \, dx \right) \Phi_{T,P} b_T$$

With n=2 and restricted to a triangle

On a given triangle T with barycentric coordinates $\lambda_1, \lambda_2, \lambda_3$ and given the averaging coefficients $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$, the bubble coefficients $\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ and the volume coefficients (c_1, c_2, c_3) the following functions are defined:

$$J_1 v_{\text{CR}} := a_1 \lambda_1 + a_2 \lambda_2 + a_3 \lambda_3 \in P_1$$

$$b_F := 6 \prod_{P \in \mathcal{V}(F)} \lambda_P \in P_2$$

$$J_2 v_{\text{CR}} := J_1 v_{\text{CR}} + 6b_1 \lambda_2 \lambda_3 + 6b_2 \lambda_1 \lambda_3 + 6b_3 \lambda_1 \lambda_2 \in P_2$$

$$\Phi_P := \frac{1}{\sqrt{|T|}} (\alpha - \beta \lambda_P) \in P_1$$

$$J_3 v_{\text{CR}} := J_2 v_{\text{CR}} + 27\lambda_1 \lambda_2 \lambda_3 (c_1 \Phi_1 + c_2 \Phi_2 + c_3 \Phi_3) \in P_4$$

Calculate $J_3 v_{\text{CR}} = J_3 v_{\text{CR}}(\lambda_1, \lambda_2, \lambda_3)$:

$$J_3 v_{\text{CR}} = J_2 v_{\text{CR}} + 27\lambda_1 \lambda_2 \lambda_3 (c_1 \Phi_1 + c_2 \Phi_2 + c_3 \Phi_3)$$

$$= J_1 v_{\text{CR}} + 6b_1 \lambda_2 \lambda_3 + 6b_2 \lambda_1 \lambda_3 + 6b_3 \lambda_1 \lambda_2 + \frac{27}{\sqrt{|T|}} \lambda_1 \lambda_2 \lambda_3 [c_1 (\alpha - \beta \lambda_1) + c_2 (\alpha - \beta \lambda_2) + c_3 (\alpha - \beta \lambda_3)]$$

$$= a_1 \lambda_1 + a_2 \lambda_2 + a_3 \lambda_3 + 6(b_1 \lambda_2 \lambda_3 + b_2 \lambda_1 \lambda_3 + b_3 \lambda_1 \lambda_2) + \frac{27}{\sqrt{|T|}} \lambda_1 \lambda_2 \lambda_3 [\alpha (c_1 + c_2 + c_3) - \beta (c_1 \lambda_1 + c_2 \lambda_2 + c_3 \lambda_3)]$$

Integrate $J_3 v_{\text{CR}}$:

Compute integrating coefficients:

using $\int_T \lambda_1^\alpha \lambda_2^\beta \lambda_3^\gamma = 2|T| \frac{\alpha! \beta! \gamma!}{(2+\alpha+\beta+\gamma)!}$ we get the following coefficients:

$$\int_T 1 = |T|$$

$$\int_T \lambda_j = \frac{1}{3}|T|$$

$$\int_T \lambda_i \lambda_j = \frac{1}{12}|T|$$

$$\int_T \lambda_i^2 = \frac{1}{6}|T|$$

$$\int_T \lambda_i \lambda_j \lambda_k = \frac{1}{60}|T|$$

$$\int_T \lambda_i^2 \lambda_j = \frac{1}{30}|T|$$

$$\int_T \lambda_i^3 = \frac{1}{10}|T|$$

$$\int_T \lambda_i^2 \lambda_j \lambda_k = \frac{1}{180}|T|$$

Exact integration on one triangle T :

$$\begin{aligned} \int_T J_3 v_{\text{CR}} &= \int_T a_1 \lambda_1 + a_2 \lambda_2 + a_3 \lambda_3 + 6(b_1 \lambda_2 \lambda_3 + b_2 \lambda_1 \lambda_3 + b_3 \lambda_1 \lambda_2) + \frac{27}{\sqrt{|T|}} \lambda_1 \lambda_2 \lambda_3 [\alpha(c_1 + c_2 + c_3) - \beta(c_1 \lambda_1 + c_2 \lambda_2 + c_3 \lambda_3)] \\ &= \frac{1}{3}|T|(a_1 + a_2 + a_3) + \frac{6}{12}|T|(b_1 + b_2 + b_3) + \frac{27\sqrt{|T|}}{60}\alpha(c_1 + c_2 + c_3) - \frac{27\sqrt{|T|}}{180}\beta(c_1 + c_2 + c_3) \\ &= |T|\left(\frac{1}{3}a_{\text{sum}} + \frac{1}{2}b_{\text{sum}}\right) + \frac{3\sqrt{|T|}}{20}(3\alpha - \beta)c_{\text{sum}} \end{aligned}$$

Exact integral on Ω :

$$\int_\Omega J_3 v_{\text{CR}} = \sum_{T \in \mathcal{T}} \int_T J_3 v_{\text{CR}}$$